

EXPANSION_PLAN_v2

Provider-Coverage Expansion Plan v2 (REWRITE) — HelixCode / HelixAgent / HelixLLM / LLMsVerifier / claude_toolkit

Date: 2026-07-08 (rewrite of the same-day v2 draft, following a NO-GO independent review) **Type:** Planning document only. No product code touched by this rewrite. No `git add/commit`. **Author:** T-planning subagent (read-only research + design). **Status:** REWRITE — the prior draft of this file was reviewed and returned **NO-GO** (scratchpad/review_providers_plan_v2.md, §11.4.142/§11.4.150) because its Phase 1 and most of its Phase 2.1 re-planned work that had **already landed, been tagged, and been declared RELEASE PUBLISHED** before the draft was written — a §11.4.74 catalogue-check failure. This rewrite corrects every Critical/Important finding from that review and is scoped as a **delta over the shipped baseline**, not a from-scratch re-plan.

Anti-bluff (§11.4.6/§11.4.99/§11.4.150): every claim below is cited either to `file:line` from an actual read this session, to a `git log/git fetch` command actually run this session, or to an external URL fetched/searched 2026-07-08 (carried forward from the reviewed draft where the review did not dispute it). Nothing is re-derived from a stale in-tree research doc without independently confirming the doc's underlying premise (registry/config state) against the live source this session.

§0 — What is ALREADY LANDED (the baseline this plan extends — READ FIRST)

This section did not exist in the reviewed draft's Phase-1 framing; the draft described this work as pending/BLOCKED. It is not. All of the following is **already committed on the current branch** (feature/helixllm-full-extension, this repo's submodules/llms_verifier HEAD), confirmed by `git log --oneline -15 + direct file reads` this session:

0.1 LLMsVerifier C1 → C5 capability-resolver chain — LANDED, not BLOCKED

submodules/llms_verifier/llm-verifier/capabilities/registry.go (read in full this session, lines 1-60+) is **already** providerCapabilitySeeds (not the old providerCapabilities map the reviewed draft described) — the doc-comment at registry.go:7-14 states verbatim: “hand-authored bootstrap DEFAULTS ONLY — they are NOT verified and MUST be overridden by a live probe ... The fail-closed resolver (ResolveModelCapability, registry_resolve.go) prefers a fresh per-model database.VerificationResult and reports unverified when none exists.” This IS the C3 fail-closed resolver the reviewed draft’s Phase 1.4 described as “BLOCKED pending C4+C5 landing.” It has landed. Commits (this session’s git log --oneline -15 on submodules/llms_verifier):

Commit	Date	What
309af635	2026-07-06 16:41	C1 — Message.ToolCalls
2c507020	2026-07-06 17:04	C2 — RAG/Skills/Plugins capability fields
09f9533c	2026-07-06 21:32	C4 — per-capability verification probes
28e6625a	2026-07-06 21:44	C5 — wire Verify
ad18e91f	2026-07-06 21:56	C3 — fail-closed resolver (RED → GREEN)
d1f04e5c	2026-07-06 22:11	review-advisory fix (detector gate + sentinel oracle)
c696c5db	2026-07-07 12:07	extended OpenAI-compatible provider config records + reachability proof (§0.2)

docs/research/07.2026/00_master/RESUME.md (this project’s own §11.4.131 session-resumption file, timestamped 2026-07-08 02:16 — **before** the reviewed draft was written) states in plain text: “LLMsVerifier chain: C1 C2 C4 C5 C3 + advisories landed, combined review GO” and that tag helix-code-1.0.0-dev-0.0.1 is **RELEASE PUBLISHED**, citing llms_verifier(c696c5db) as one of the published submodule SHAs. **This plan does not re-propose the C1–C5 chain in any form.**

0.2 Extended-provider config rows — LANDED with partial live proof, not pending

submodules/llms_verifier/llm-verifier/providers/config.go (grep run this session, registerDefaultProviders(), 1396 lines) already registers **13 provider config rows** beyond the long-standing core set — confirmed by direct grep for pr.providers["<id>"] =:

poe (1086), perplexity (1109), sakana (1132), hunyuan (1155), xai (1179), moonshot (1203), zai (1228), fireworks (1251), deepinfra (1274), ai21 (1297), reka (1321) — plus pre-existing hyperbolic (687, real endpoint https://api.hyperbolic.xyz/v1, NOT a stub) and xiaomi (1028). **Baseten is genuinely absent** (confirmed — no pr.providers["baseten"] hit).

Commit c696c5db (2026-07-07, one day before the reviewed draft) — “feat(providers): extended OpenAI-compatible provider config records + reachability proof (CONST-036/§11.4.28)” — landed these rows with a live reachability test (providers/extended_providers_test.go) that captured REAL evidence at docs/qa/phase4_extended_providers_20260707/: **poe LIVE-PROVEN (341 models), zai LIVE-PROVEN (8 models), novita LIVE-PROVEN (143 models), fireworks reachable-but-billing- blocked (HTTP 412 → honest SKIP), 9 credential-absent honest SKIPS.**

This plan does not re-propose adding these 13 config rows. The genuinely remaining work is narrower: (a) extend the live-proof coverage from 3 fully-proven + 1 partial to the remaining ~9 rows once API keys are available (operator-gated, §11.4.10 — tracked in §3.4 below, not scriptable today); (b) reconcile the **separate, sibling** claude_toolkit repo’s own alias/key-routing files against this list (§2 below — that file lives in a different repo and was NOT touched by c696c5db).

0.3 Operator-resolved clarifications — already answered, not open questions

docs/research/07.2026/00_master/03_open_clarifications.md (created 2026-07-06, last updated 2026-07-08 — same day as, but before, the reviewed draft) already records **operator-resolved** answers for the four items the reviewed draft re-derived independently:

Item	Operator resolution (already recorded, 03_open_clarifications.md C1–C4)
GPT-Sol	

Item	Operator resolution (already recorded, 03_open_clarifications.md C1–C4)
	OpenAI GPT-5.6 “Sol” — no new adapter; add model IDs to the existing OpenAI adapter at GA
Google OKF	RAG/Skills knowledge format via MCP resources server — NOT a provider, out of scope
ACP	Google A2A (agent ↔ agent) — NOT Zed ACP; a protocol thread, not a provider
Subquadratic	BLOCKED-until-GA — no public base URL/auth; no adapter work possible

PROVIDER_COVERAGE.md (same directory) independently arrives at matching conclusions via its own research pass and cites the same four items — this rewrite treats 03_open_clarifications.md as the operator-authoritative record and does not re-litigate any of the four (§11.4.74 — the reviewed draft’s independent re-derivation of the same four answers was itself flagged by the review as duplicated effort; not repeated here).

§1 — Corrections applied per the NO-GO review (traceability)

Review finding	Correction applied in this rewrite
C1 — Phase 1 (C1 → C5) presented as pending/BLOCKED	§0.1 above states plainly it is LANDED, cites the same 6 commits the review found, and the phased plan below (§3) contains zero C1–C5 re-implementation tasks
C2 — Phase 2.1 (Poe/Sakana/Hunyuan registration) duplicated c696c5db’s already-landed config rows; PROVIDER_COVERAGE.md/ 03_open_clarifications.md operator resolutions not discovered/ cited	§0.2 cites c696c5db + the exact config.go line numbers + the captured evidence path; §0.3 cites the operator-resolved four items instead of re-deriving them; §2 below reconciles ONLY the claude_toolkit-side gap that genuinely remains (sakana/hunyuan/

Review finding	Correction applied in this rewrite
	moonshot/fireworks/deepinfra/ai21/reka aliases — checked this session, see §2.1)
C3 — claude_toolkit’s vendored LLMsVerifier freshness claim made with no <code>git fetch</code>; wrong target branch for Phase 7	Ran <code>git fetch --all --prune</code> in <code>claude_toolkit</code> this session (read-only) — confirmed <code>claude_toolkit</code>’s own <code>origin/main</code> has NO new commits (HEAD <code>..origin/main</code> empty, HEAD <code>9d12347</code>). Ran <code>git fetch origin --prune --tags</code> inside <code>claude_toolkit/submodules/LLMsVerifier</code> this session — confirmed local <code>17b4bfb6</code> is now 3 commits behind that remote’s <code>origin/main</code> (<code>ae41f5a0</code>), AND confirmed via <code>git merge-base --is-ancestor c696c5db origin/main</code> → NO — <code>c696c5db</code> lives on <code>origin/feature/helixllm-full-extension</code>, not <code>origin/main</code>, on the SAME upstream (<code>vasic-digital/LLMsVerifier</code>) both this meta-repo’s <code>submodules/llms_verifier</code> and <code>claude_toolkit</code>’s vendored copy point at. §3.5 below re-scopes the Phase-7-equivalent task honestly: it is blocked on an operator-approved merge of feature/helixllm-full-extension → main (§11.4.167 — no trunk merge without explicit approval), not a script that can run today against <code>origin/main</code>
I1 — every <code>helix_code/helix_code/...</code> citation is a doubled, non-existent path	Confirmed this session: <code>ls helix_code/helix_code</code> → no such directory; the real paths are <code>helix_code/internal/server/*.go</code> and <code>helix_code/internal/llm/openai_provider.go</code> (single level, per this project’s own CLAUDE.md §3.2.1). Every citation below uses the corrected single-level path. The underlying finding

Review finding	Correction applied in this rewrite
	(HelixCode's own server registers no /v1/chat/completions or /v1/messages route) is unchanged — re-confirmed by <code>grep -rn "chat/completions\ /v1/messages\ router.POST" helix_code/internal/server/*.go</code> this session, zero hits
I2 — “Xiaomi dual OAI+Anthropic” conflated vendor API with local adapter	Confirmed this session (grep on <code>submodules/helix_agent/internal/llm/providers/xiaomi/xiaomi.go</code> , 605 lines): the local adapter implements only <code>https://api.xiaomimimo.com/v1/chat/completions</code> (OpenAI-style) — zero occurrences of <code>/v1/messages</code> , <code>anthropic-version</code> , or <code>x-api-key</code> . Xiaomi's <i>vendor</i> API additionally documents an Anthropic-compatible surface (<code>03_open_clarifications.md</code>), but the local HelixAgent adapter does not implement or need it — the “already-wired” classification for Xiaomi stands (the dedicated adapter is real, substantial code), the “dual wire” framing is now scoped correctly as describing the external vendor, not local code

Preserved unchanged (review confirmed these accurate — not re-derived, not altered): the 1-net-new-adapter conclusion (dual OpenAI+Anthropic wire facade for HelixCode's own server, §3.1 below); the generic/generic.go:27-30 config-only seam; claude_toolkit's PATH-detection implementation state + the R5 port-ownership hardening as a genuinely-open gap (confirmed this session — `detect_helixagent_record()` at `claude-providers.sh:178` - accepts *any* non-empty `data[].id` list via `jq -r '.data[].id? // empty'` rather than asserting the known helix-debate/helix-llm ids are present, lines 185-186/207/210); GPT-Sol and Subquadratic held at UNCONFIRMED-needs - endpoint (re-searched 2026-07-08, still no public GA/base-URL — see the reviewed draft's Step-2 citations, not disputed by the review and not re-searched

again here to avoid duplicate WebSearch spend); §11.4.6 honesty markers throughout.

§2 — claude_toolkit alias reconciliation (the genuinely-remaining slice of the old Phase 2.1)

Checked this session (cat scripts/providers/key-aliases.json scripts/providers/overrides.json in claude_toolkit, a **separate sibling repo** at /home/milos/Factory/projects/tools_and_research/claude_toolkit — NOT touched by c696c5db, which only landed in submodules/llms_verifier's config.go):

Provider (from §0.2's landed config.go list)	Already in claude_toolkit's own alias files?	Action needed
poe	YES — POE_API_KEY→poe (+ ApiKey_Poe) in key-aliases.json; poe.strong_model/fast_model pinned in overrides.json	None
xai	YES — xai.base_url pinned in overrides.json	None
zai	YES — as zai-coding-plan (ZAI_API_KEY→zai-coding-plan)	None
xiaomi	YES — XIAOMI_MIMO_API_KEY→xiaomi, native transport + base_url pinned	None
hunyuan	PARTIAL/AMBIGUOUS — TENCENT_CLOUD_API_KEY→tencent-tokenhub exists, but there is no hunyuan alias and tencent-tokenhub is not confirmed to be the same endpoint as https://api.hunyuan.cloud.tencent.com/v1	Genuinely open — reconcile whether tencent-tokenhub already targets Hunyuan or is a distinct Tencent product, before adding a possibly-duplicate hunyuan alias (§11.4.122 no-silent-removal applies in reverse:

Provider (from §0.2's landed config.go list)	Already in claude_toolkit's own alias files?	Action needed
		don't silently ADD a duplicate either)
sakana	NO	Genuinely open — no existing alias/override; safe to add
moonshot	AMBIGUOUS — kimi-for-coding exists with base_url: https://api.kimi.com/coding/, distinct from PROVIDER_COVERAGE.md's https://api.moonshot.ai/v1 general-chat endpoint; config.go's own comment at line 1202 notes moonshot is "distinct from the pre-existing 'kimi' record (api.moonshot.cn, China endpoint)"	Genuinely open — three possibly-distinct Kimi/Moonshot surfaces (China kimi.cn, international moonshot.ai, coding-specific kimi.com/coding) need reconciling before adding an alias, not a blind add
fireworks, deepinfra, ai21, reka	NO	Genuinely open — no existing alias/override; safe to add once operator confirms priority (these are lower-salience per the original Wave-2 rollout ordering, which the review did not dispute)

Corrected task (replaces the old Phase 2.1): a pre-flight reconciliation pass — NOT a blind “add three aliases” task — resolving the hunyuan/tencent-tokenhub and moonshot/kimi-for-coding ambiguities via a live endpoint comparison (does tencent-tokenhub's pinned base_url, if any, resolve to the Hunyuan API?) before any new alias is written, then adding sakana (unambiguous) + the four lower-priority providers (fireworks/deepinfra/ai21/reka) with the SAME transport: router convention already used for HelixAgent (OpenAI-shaped, not native).

§3 — Phased plan (genuinely-remaining work only)

Every phase below is a **delta task** — nothing here re-implements §0's landed baseline.

Phase A — HelixLLM-as-tracked-provider registration in LLMsVerifier (still open)

This is the ONE piece of the old Phase 1 that was genuinely NOT yet landed (the C1–C5 chain is a generic capability-resolver mechanism; HelixLLM itself registering as a tracked provider instance is a separate, still-open task). `submodules/helix_llm/README.md` (read this session) confirms HelixLLM exposes both OpenAI-compatible (`/v1/chat/completions`, `/v1/models`) and Anthropic-compatible (`/v1/messages`) surfaces on one server (default port 8443 TLS; proven-live plain-HTTP llama-server sidecar at :18434 per `RESUME.md`'s live-server section — base URL WITHOUT `/v1` per the corrected `PHASE2_BLOCKERS_INVESTIGATION.md` OQ1 finding).

A.1 Register a `helixllm ProviderDescriptor` in the now-landed seed/resolver mechanism (`registry.go` seeds + `registry_resolve.go`'s fail-closed resolver), sourcing `BaseURL` from `HELIX_LLM_LOCAL_OPENAI_ENDPOINT` (no `/v1` suffix), `Wire: "openai"`, health path `/internal/health`. **A.2** If the landed resolver supports only one wire per logical provider ID, register a second ID (e.g. `helixllm-anthropic`) pointed at the same base URL with `Wire: "anthropic"` for the `/v1/messages` surface — confirm against the landed `registry_resolve.go` implementation (genuinely not yet checked this session — a real open sub-task, not carried forward as a guess). **A.3** Acceptance: a real `curl {endpoint}/v1/models` (or the resolved endpoint) non-empty data[], captured to `docs/qa/<run-id>/helixllm_verifier_registration/models.json`, plus a real `VerificationResult` row for `provider="helixllm"` from the landed C4 probe mechanism — `Responsive=true`, at least one `Supports*` flag backed by a real probe artifact, never a config-only “registered” claim.

Phase B — claude_toolkit alias reconciliation (§2 above)

B.1 Resolve the tencent-tokenhub vs hunyuan ambiguity and the kimi-for-coding vs moonshot ambiguity by comparing live-resolved base URLs (§2 table) — a one-command check, not a design change. **B.2** Add sakana (unambiguous, no existing alias) + fireworks/deepinfra/ai21/reka (lower priority, add after B.1 resolves) to `key-aliases.json/overrides.json` with `transport: router` (OpenAI-shaped). **B.3** Acceptance: `claude-providers sync` with real keys produces new `<id>.env` files with correct `CMA_PROVIDER_BASE_URL/TRANSPORT`; `scripts/tests/verify_providers_live.sh` produces `proof/<n>-sakana-`

live.txt etc. showing a real, non-simulated round-trip — mirroring the existing evidence pattern already in that repo.

Phase C — claude_toolkit HelixAgent-provider R5 port-ownership hardening (still open, confirmed)

C.1 `detect_helixagent_record()` (`claude-providers.sh:178-`) currently accepts any non-empty `data[].id` list from a `:8100` responder. Harden it to assert the returned id set contains `helix-debate` OR `helix-llm` (the `CMA_HELIXAGENT_STRONG/_FAST` env-overridable defaults already declared at lines 185-186) before treating the port as HelixAgent's — prevents a false-positive registration against an unrelated service coincidentally listening on the same port (a §11.4.174-class port-ownership hazard). **C.2** Acceptance: (1) real `/v1/models` response containing the known ids, captured; (2) a NEGATIVE-case test proving a fake responder returning `{"data": []}` or an unrelated JSON shape does NOT get registered as the `helixagent` alias.

Phase D — Dual OpenAI-style + Anthropic-style wire facade for HelixCode's own API (the one net-new adapter — unchanged from the reviewed draft's Phase 5, review-confirmed accurate)

Confirmed gap, corrected paths (I1): `grep -rn "chat/completions\|/v1/messages\|router.POST" helix_code/internal/server/*.go` this session — zero hits; HelixCode's own server (`server.go`, `handlers.go`, `llm_generate.go`, `qa_handlers.go`, `specify.go`) registers only its own custom `/api/v1/llm/generate` shape, never serves an OpenAI or Anthropic wire to external callers.

D.1 Read HelixLLM's own dual-facade implementation (not yet read this session — first sub-task) as the worked precedent, per the fork points already enumerated in the reviewed draft's R4 (request-shape system-field handling, response-shape content-block-array-vs-flat-string, SSE-framing divergence, tool-calling schema translation, auth-header divergence) — none of that enumeration was disputed by the review, carried forward unchanged. **D.2** Design a shared internal `GenerateRequest/GenerateResult` extending `helix_code/internal/server/llm_generate.go`'s existing `generate/stream` core. **D.3** Add `POST /v1/chat/completions` + `POST /v1/messages` route groups to `helix_code/internal/server/server.go`, both delegating to the shared core, with two distinct SSE encoders. **D.4** Bidirectional tool-calling schema translation (JSON-Schema wrapper-key difference only). **D.5** Register both routes with the landed `LLMsVerifier` resolver as a `helixcode/helixcode-anthropic` provider pair (mirrors Phase A.2). **D.6** Acceptance: real curl round-trips on both endpoints (non-empty `choices[0].message.content / content[0].text`), streaming variants

with correct SSE framing, a tool-call round-trip on both shapes — all captured to docs/qa/<run-id>/dual_wire_facade/, no metadata-only pass accepted.

Phase E — Setup/PATH-install script gap-fill (small, unchanged from reviewed draft §1.7/Phase 4 — not disputed)

install_helix_path.sh (304 lines, read in full in the reviewed draft's session) already builds+installs all four binaries (helixcode, helixagent, helixllm, llms-verifier), no sudo, idempotent rc-patching, black-box --version probe per binary. The review did not dispute this. Remaining gaps:

E.1 (optional) A fifth component row symlinking claude_toolkit's own CLI entry point, compliant with §11.4.177 (the symlink target resolves its own SCRIPT_DIR, never a copy). **E.2** Extend the existing black-box test with a helixllm --help | grep -q '\-\-mode' sanity assertion (confirms the mode-flag surface the README documents is actually present in the built binary) — one additional assertion within the existing philosophy, not a new mechanism.

Phase F — Provider-verification test extension (scoped to what Phase A/B actually add)

F.1 Confirm whether verify_providers_live.sh's provider-iteration loop is already env-file-driven (and therefore automatically picks up new <id>.env files from Phase B) or has a hardcoded provider-id list needing the new ids appended — a read-only check, not yet performed this session. **F.2** Extend test_providers.sh's hermetic fixture set with the Phase-B-added providers' fake catalog entries, so the resolver's classification logic is exercised without live network calls. **F.3** Once keys are available (operator-gated, §11.4.10), extend the c696c5db live-proof harness (providers/extended_providers_test.go) from its current 3-fully-proven-of-13 state toward full coverage of the already-registered §0.2 config rows — this is NOT new adapter work, it is closing the evidence gap the tag's own §11.4.138 honesty-correction entry (docs/research/07.2026/00_master/PROVIDER_COVERAGE.md, bottom section) already tracks as a named follow-up.

Phase G — Operator-gated / non-actionable-today items (tracked, not scheduled)

G.1 claude_toolkit's vendored submodules/LLMsVerifier bump — per §1's C3 correction, this is **honestly blocked**, not scriptable today: the C1–C5 + extended-provider commits live on origin/feature/helixllm-full-extension of vasic-digital/LLMsVerifier, not origin/main. Two honest paths, either requiring an explicit operator decision (§11.4.66): (a) wait for feature/helixllm-full-extension → main merge (operator-approved per §11.4.167),

then run the existing `11_claude_toolkit.md` §2.3 fetch/merge/rebuild procedure against the now-correct `origin/main`; or (b) if the operator wants the extended-provider work in `claude_toolkit` sooner, point its vendored submodule at the feature branch explicitly (a deviation from the “always track main” convention that needs its own operator sign-off). No script is authored for either path until the operator picks one — scripting path (a) against today’s `origin/main` would reproduce exactly the bug the review caught (Phase 7’s own acceptance criterion would fail on a live run, as the review demonstrated). **G.2** Qwythos-9B training-data-provenance operator sign-off (`dev.ua`’s “trained on Claude-distilled data” finding) — unchanged from the reviewed draft, not yet asked. **G.3** GPT-Sol / Subquadratic — held at UNCONFIRMED-needs-endpoint / BLOCKED-until-GA, per `03_open_clarifications.md`’s already-operator-resolved verdicts (§0.3). No adapter work; revisit only on a GA/public-API announcement. **G.4** Baseten — confirmed genuinely absent from `config.go` (unlike Hyperbolic, which already has a real config row per §0.2). Per-deployment base URL, no single global endpoint — needs a concrete deployment URL injected before any work, per the reviewed draft’s original §1.2 note (not disputed by the review).

§4 — Net-new-work count (post-reconciliation)

Zero re-implementation of the landed C1–C5 chain or the 13 landed config rows. **Genuinely remaining, actionable-today items: 6** (Phase A HelixLLM-registration, Phase B alias reconciliation, Phase C R5 hardening, Phase D dual-wire facade — the one net-new adapter, Phase E setup-script gap-fill, Phase F test extension). **Operator-gated, not scriptable today: 4** (Phase G.1–G.4). This plan does not re-plan any shipped work; every task above is either newly-discovered this session (Phase A.2’s resolver-capacity question, §2’s alias ambiguities) or a review-confirmed still-open gap (Phase C, Phase D, Phase E).

Sources verified 2026-07-08

In-tree source read/grepped THIS rewrite session (facts, no guessing — §11.4.6): - `submodules/llms_verifier/llm-verifier/capabilities/registry.go:1-60` (full seed-map header + first provider block) - `submodules/llms_verifier/llm-verifier/providers/config.go` (full `pr.providers["<id>"] = grep`, 1396 lines) - `git -C submodules/llms_verifier log --oneline -15,git -C submodules/llms_verifier branch --show-current,git -C submodules/llms_verifier remote -v - docs/research/07.2026/00_master/RESUME.md` (full) - `docs/research/07.2026/00_master/PROVIDER_COVERAGE.md` (full) - `docs/research/07.2026/00_master/03_open_clarifications.md` (full) - `helix_code/`

```
internal/server/ directory listing + grep -rn "chat/completions\|/v1/
messages\| router.POST" helix_code/internal/server/*.go (zero hits,
corrected single-level path per I1) - submodules/helix_agent/internal/llm/
providers/xiaomi/xiaomi.go (grep for wire-shape markers, 605 lines total —
confirmed OpenAI-only) - /home/milos/Factory/projects/
tools_and_research/claude_toolkit: git fetch --all --prune (read-
only), git log --oneline -1 (HEAD 9d12347), git log --oneline
HEAD..origin/main (empty) - claude_toolkit/submodules/LLMsVerifier:
git fetch origin --prune --tags (read-only), git log --oneline -1
(local 17b4bfb6), git log --oneline origin/main -1 (ae41f5a0), git
merge-base --is-ancestor c696c5db origin/main → NO, git branch -r
--contains c696c5db → origin/feature/helixllm-full-extension -
claude_toolkit/scripts/providers/key-aliases.json + overrides.json
(full contents, this session) - claude_toolkit/scripts/claude-providers.sh
(grep for detect_helixagent_record, data[].id, helix-debate/helix-llm
— lines 178-210)
```

Carried forward, NOT re-searched this rewrite session (the review did not dispute these; re- searching would be duplicate WebSearch spend against a same-day finding): Sakana Fugu vendor- confirmed OpenAI-compatible wire shape, Subquadratic private-beta status, Google OKF / MCP-resources classification, GPT-5.6 “Sol” limited-preview status, Qwythos-9B self-host + provenance flag — all originally cited in the reviewed draft’s Step-2 table with URLs accessed 2026-07-08, unchanged by this rewrite’s corrections (which are structural/state corrections, not provider-identity corrections).

Review this rewrite responds to: scratchpad/
review_providers_plan_v2.md (NO-GO, Critical C1/C2/C3, Important I1/I2 — all addressed above; see §1 traceability table).